

Day 4 Student Handout

Geometry Summer Credit | Rigid Motions, Congruence, and Triangle Angles

Name:	Date:	Period:
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Learning Targets

I can identify rigid motions and use them to understand congruence. I can classify triangles and use interior and exterior angle relationships to find missing measures.

Lesson 4A: Rigid Motion and Congruence

Warm-Up

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| 1. | Translate A(-2, 5) by $\langle 6, -3 \rangle$. Then reflect B(4, -7) across the y-axis. |
| 2. | Rotate C(-3, 2) 180 degrees about the origin. Dilate D(2, -6) by scale factor 3. |

Concept Launch

Same Size, Same Shape

Translations, reflections, and rotations are rigid motions. They preserve distance and angle measure. A dilation is not a rigid motion unless the scale factor is 1.

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| 3. | Draw any triangle on graph paper. Translate it, reflect it, and rotate it. Which measurements appear to stay the same each time? |
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Guided Examples

Problem	Work / notes
Triangle ABC has side lengths 5, 7, and 9. Triangle A'B'C' is a translation of ABC. What are the side lengths of A'B'C'?	Rigid motion? Side lengths:
A figure is reflected across the x-axis. Does the image stay congruent to the original? Explain.	What changes? What stays the same?
Triangle PQR maps to triangle XYZ by a rotation.	Corresponding angles:

If angle $P = 48$ degrees, angle $Q = 67$ degrees, and $PQ = 10$, name corresponding parts in the image.	Corresponding side:
A triangle is dilated by scale factor 2. Is the image congruent to the original? Why or why not?	Same shape? Same size? Congruent?

Practice

4.	Name three rigid motions.
5.	A triangle is rotated 90 degrees. Are the original and image congruent? Explain.
6.	Triangle ABC is congruent to triangle DEF. If $AB = 8$, $BC = 11$, and $AC = 13$, what are DE, EF, and DF?
7.	Error analysis: A student says a dilation by scale factor 3 creates a congruent figure because the shape looks the same. Explain the mistake.

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Lesson 4B | Triangle Classification and Angle Sums

Lesson 4B: Triangle Classification and Angle Relationships

Warm-Up

1.	Find the missing angle in a triangle with angles 42 degrees and 68 degrees.
2.	Solve $3x + 18 = 96$ and $5x - 7 = 2x + 38$.

Key Ideas

Triangle Facts Interior angles of a triangle sum to 180 degrees. An exterior angle equals the sum of the two remote interior angles. Triangles can be classified by sides and by angles.
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Guided Examples

Problem	Work / notes
Classify a triangle with side lengths 6, 6, and 10.	By sides: Reason:
Classify a triangle with angles 35 degrees, 55 degrees, and 90 degrees.	By angles: Reason:
Find x if triangle angles are x, 2x, and 3x.	Equation: Solve: Angles:
An exterior angle is 124 degrees. One remote interior angle is 49 degrees. Find the other remote interior angle.	Equation: Missing angle:

Practice

3.	Classify by sides: 7, 7, 7.
4.	Classify by angles: 40 degrees, 60 degrees, 80 degrees.

5. Find the missing angle: 31 degrees, 82 degrees, and x .

6. Triangle angles are $4x$, $5x$, and $6x$. Find x and all three angles.

7. An exterior angle is 132 degrees. One remote interior angle is 57 degrees. Find the other remote interior angle.

8. Reflection: Which triangle relationship felt most useful today?